

Human COVID-19 Immune scRNA-seq Analysis

Single-cell transcriptomic analysis of immune-cell populations from COVID-19 and healthy samples

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1. Project Overview

This project analyzes human single-cell RNA sequencing (scRNA-seq) data from immune-cell samples from COVID-19 and healthy individuals.

The analysis includes quality control, doublet detection, normalization, highly variable gene selection, sample integration, dimensionality reduction, clustering, cell-type annotation, cell-type composition analysis, differential expression analysis, and pathway enrichment.

Analysis Workflow

Expression Matrices → Quality Control → Doublet Detection → Normalization → Highly Variable Gene Selection → Batch Correction → PCA → UMAP → Clustering → Cell-Type Annotation → Cell-Type Proportions → Differential Expression → GO Pathway Enrichment

2. Dataset and Sample Metadata

The dataset contains four immune-cell samples:

Sample	Condition
covid_556	COVID-19
covid_557	COVID-19
covid_558	COVID-19

HIP043	Healthy
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The initial expression matrices contained:

Sample	Cells	Genes
covid_556	6,956	27,260
covid_557	15,102	29,509
covid_558	18,182	29,472
HIP043	11,943	29,205

The Seurat metadata included sample identity, total RNA counts, detected genes, donor, replicate, and sex.

3. Quality Control and Preprocessing

Quality control was performed separately for each sample before merging.

Three QC metrics were used:

- **nFeature_RNA:** number of detected genes per cell
- **nCount_RNA:** total RNA counts per cell
- **percent.mt:** percentage of mitochondrial RNA

Cells with fewer than 200 detected genes were removed as low-quality cells or empty droplets.

Cells with more than 6,000 detected genes were removed as potential doublets or multiplets.

Cells with more than 10% mitochondrial RNA were removed because high mitochondrial content can indicate stressed or damaged cells.

Doublet Detection

Doublet removal was performed using **DoubletFinder**. Doublets can occur when two cells are captured in the same droplet, producing an artificial expression profile containing signals from multiple cell types.

The optimal pK values identified for the four samples were:

Sample	Optimal pK
covid_556	0.08
covid_557	0.09
covid_558	0.30
HIP043	0.30

4. Normalization and Feature Selection

Normalization was performed using Seurat's default **LogNormalize** method.

For each cell, gene counts were normalized by the total number of counts in that cell, multiplied by a scale factor, and log-transformed.

Highly variable genes (HVGs) were selected to focus downstream analyses on genes showing meaningful variation across cells rather than predominantly technical variation.

HVG selection was performed using Seurat's **FindVariableFeatures()** function with the **vst** method.

5. Batch Correction and Sample Integration

The samples were compared before and after Seurat-based integration.

Before batch correction, the UMAP showed strong sample-associated separation, with cells from the four samples occupying distinct regions.

After integration, cells from different samples showed substantially improved mixing across the embedding.

This indicates that sample-associated variation contributed strongly to the original embedding and that integration reduced this variation before downstream analysis.

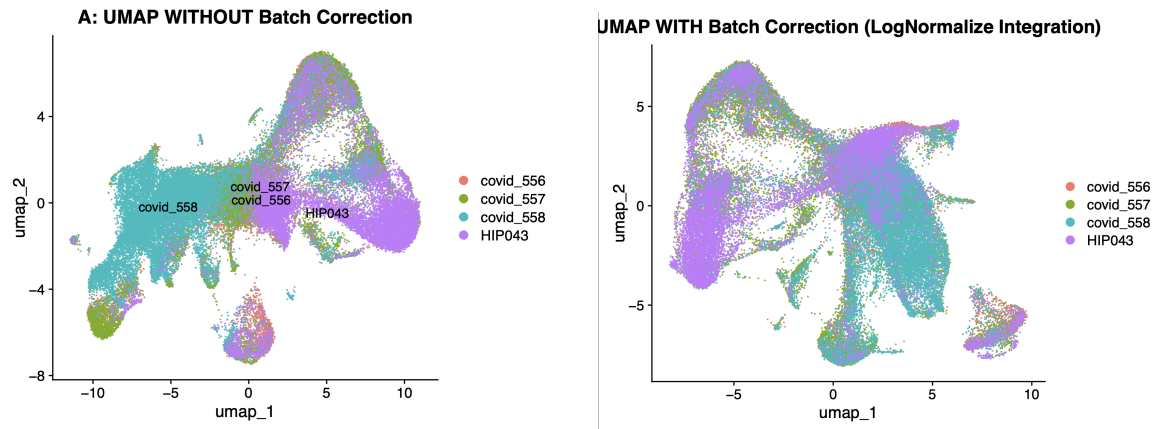


Figure 1. Comparison of UMAP embeddings before and after Seurat-based sample integration. The unintegrated embedding shows stronger sample-associated separation, while the integrated embedding shows improved mixing between samples.

6. Principal Component Analysis and UMAP

Principal Component Analysis (PCA) was used to reduce the dimensionality of the gene-expression data before downstream visualization and clustering.

The PCA representation showed cells from all four samples distributed across the principal-component space, with substantial overlap between samples.

UMAP was subsequently used to visualize the low-dimensional representation of the integrated dataset.

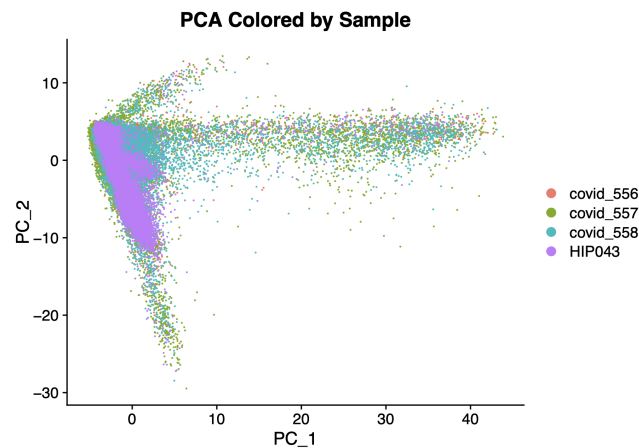


Figure 2. PCA representation of the dataset colored by sample identity.

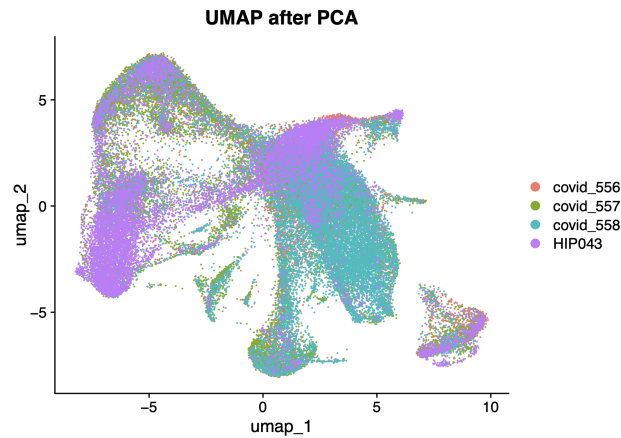


Figure 3. UMAP representation of the integrated single-cell dataset after PCA-based dimensionality reduction.

7. Cell-Type Annotation

Cell types were assigned using two complementary approaches.

Automatic Annotation

SingleR was used for automatic cell-type annotation with the **Human Primary Cell Atlas** reference.

Manual Annotation

Manual annotation was performed using cell-type marker gene expression.

The identified cell populations included:

- B cells
- T cells
- NK cells
- Plasma cells
- CD16 monocytes
- Plasmacytoid dendritic cells (pDC)
- Common lymphoid progenitors (CLP)
- Granulocyte-monocyte progenitors (GMP)
- Erythroblasts

The manual annotation identified distinct populations across the UMAP, including B cells, T cells, NK cells, plasma cells, CD16 monocytes, pDCs, CLPs, GMPs, and erythroblasts.

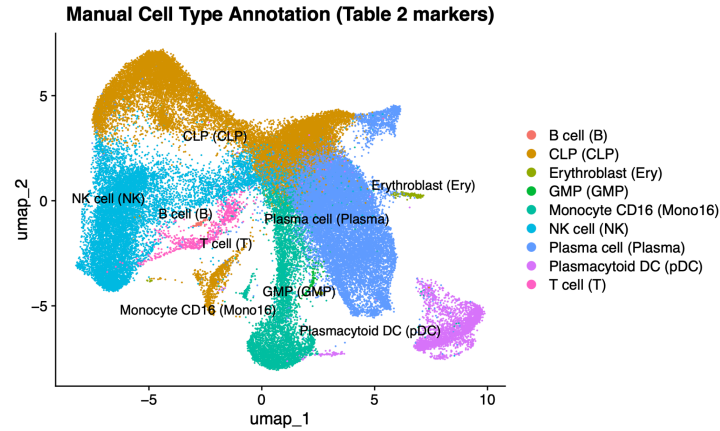


Figure 4. Manual cell-type annotation based on marker-gene expression. Distinct immune-cell populations are shown across the UMAP embedding.

8. Differential Expression Analysis

Differential expression analysis was performed between selected immune-cell populations.

The main cell-type comparisons included:

- **B cells vs T cells**
- **T cells vs CD16 monocytes**

Volcano plots were used to visualize changes in gene expression and statistical significance.

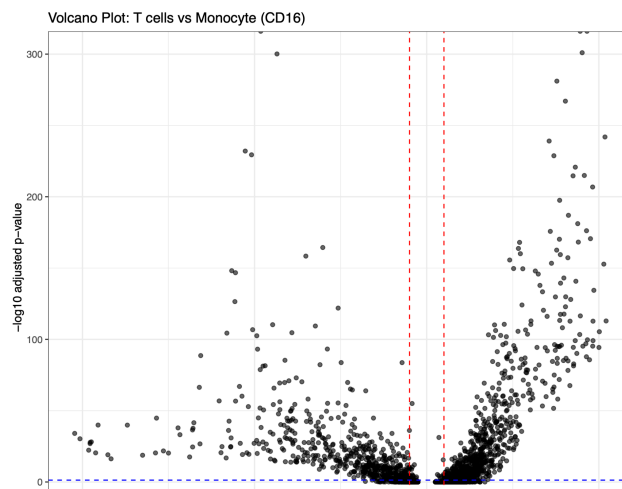


Figure 5. Volcano plot showing differentially expressed genes between T cells and CD16 monocytes. The x-axis represents log2 fold change and the y-axis represents $-\log_{10}$ adjusted p-value.

9. Top Differentially Expressed Genes

The five most significant differentially expressed genes identified in the T-cell versus B-cell comparison were:

Gene	log2 Fold Change	Adjusted p-value
RP5-1028K7.2	-7.04	8.83×10^{-73}
RASD2	-11.98	1.55×10^{-71}
CDK6	-3.58	2.25×10^{-71}
LAPTM4B	-7.48	7.75×10^{-66}
SOX4	-4.54	1.87×10^{-65}

The top differentially expressed genes were also visualized across the B-cell versus T-cell and T-cell versus CD16 monocyte comparisons.

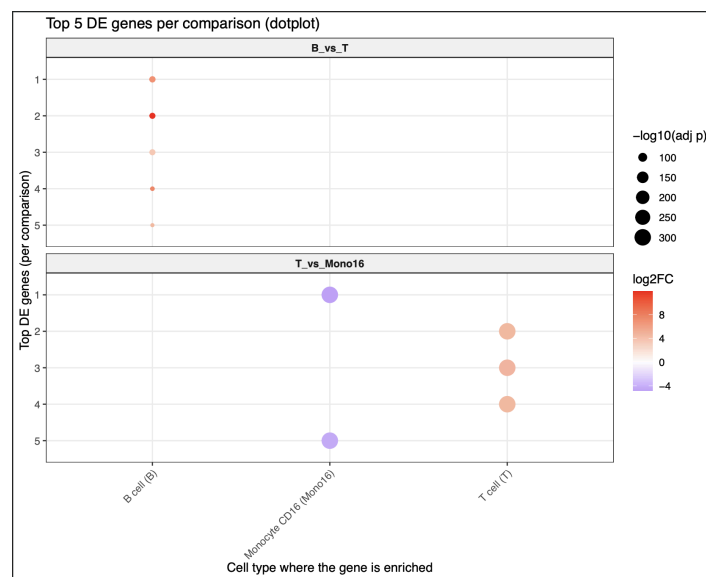


Figure 6. Dot plot showing the top differentially expressed genes identified for the B-cell versus T-cell and T-cell versus CD16 monocyte comparisons. Dot size represents statistical significance and color represents log2 fold change.

10. COVID-19 vs Healthy Differential Expression

Differential expression was also performed between COVID-19 and healthy samples.

Genes showing increased expression in the COVID-19 group included:

- IGHG4
- IGHG1
- IFI27
- S100A8
- S100A9
- LYZ
- VCAN
- IGHG3
- IGHG2
- IGHM

Genes showing decreased expression included:

- SH2D1B
- PRR20C
- SPON2
- KLRB1
- GNLY
- SLC38A7
- TRDC
- S1PR5
- MATK
- FCGR3A

The observed differences included interferon-responsive, inflammatory myeloid, immunoglobulin-associated, and cytotoxic/NK/T-cell-associated genes.

11. GO Biological Process Enrichment

Pathway analysis was performed for the T-cell versus B-cell comparison.

The pathway with the lowest p-value was:

Cytoplasmic Translation (GO:0002181)

Cytoplasmic translation represents protein synthesis by cytoplasmic ribosomes and involves ribosomal components and translation-associated factors.

Other enriched biological processes included:

- Peptide Biosynthetic Process
- Macromolecule Biosynthetic Process
- Translation
- Gene Expression
- Ribonucleoprotein Complex Biogenesis
- Ribosomal Small Subunit Biogenesis
- Ribosome Biogenesis
- Ribosome Assembly
- rRNA Metabolic Process

The enrichment results were strongly dominated by translation- and ribosome-associated biological processes.

Pathway analysis for GO terms B vs T.pdf

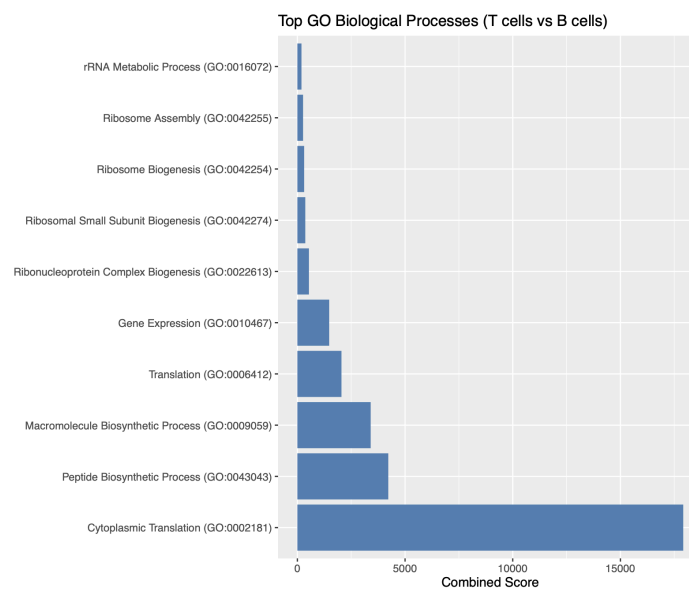


Figure 7. Top GO Biological Process terms identified for the T-cell versus B-cell comparison. Cytoplasmic Translation showed the highest combined enrichment score among the displayed terms.

12. Key Findings

- Quality control and doublet detection were used to obtain high-quality single-cell profiles.
- Sample-associated variation was reduced following Seurat-based integration.
- PCA and UMAP provided low-dimensional representations of the integrated single-cell dataset.
- Distinct immune-cell populations were identified through marker-based cell-type annotation.

- The analysis identified B cells, T cells, NK cells, plasma cells, CD16 monocytes, pDCs, CLPs, GMPs, and erythroblasts.
 - Differential expression analysis identified distinct transcriptional programs between immune-cell populations.
 - The COVID-19 versus healthy comparison showed differences in interferon-responsive, inflammatory, immunoglobulin-associated, and cytotoxic/NK/T-cell-associated genes.
 - GO enrichment identified strong representation of translation- and ribosome-associated biological processes, with Cytoplasmic Translation showing the highest combined enrichment score among the displayed terms.
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13. Tools and Methods

Programming: R
Single-cell analysis: Seurat
Doublet detection: DoubletFinder
Normalization: LogNormalize
Feature selection: Highly variable genes (vst)
Batch correction: Seurat integration
Dimensionality reduction: PCA, UMAP
Clustering: Leiden, Louvain
Cell-type annotation: SingleR, marker-based annotation
Reference: Human Primary Cell Atlas
Differential expression: Seurat
Pathway enrichment: Enrichr / GO Biological Process

14. Reproducibility

The analysis was implemented in R using a Seurat-based single-cell RNA-seq workflow.

The repository contains the analysis script, selected analysis figures, analysis results, and this project report.

15. References

Wilk, A. J. et al. (2020). Multi-omic profiling reveals widespread dysregulation of immune cell functions in COVID-19 patients.

Seurat: Satija Lab single-cell analysis framework.

DoubletFinder: Doublet detection for single-cell RNA-seq data.

SingleR: Automated cell-type annotation using reference transcriptomes.

Enrichr: Gene-set enrichment analysis platform.